Example 5.6 (Illustrating AA Similarity)

In the Fig. 5.17 , $\angle ABC \equiv \angle EDC$ and the perimeter of ΔCDE is 27 *units*, prove that $AB \equiv EC$.

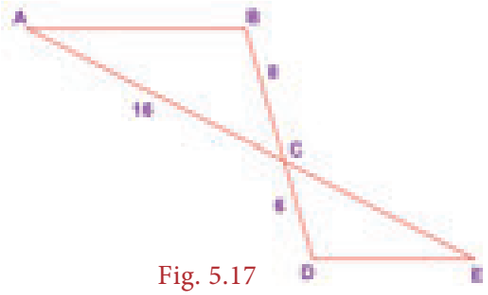


Fig. 5.17

Proof:

	Statements	Reasons
1	$\angle ABC \equiv \angle EDC$	given
2	$\angle BCA \equiv \angle DCE$	vertically opposite angles are equal
3	$\Delta ABC \sim \Delta EDC$	by AA property (1, 2)
4	$\frac{AB}{ED} = \frac{BC}{DC} = \frac{AC}{EC}$	corresponding sides are proportional by 3

$\Rightarrow \frac{8}{6} = \frac{16}{EC} \Rightarrow EC = 12$ *units*

Given, the perimeter of $\Delta CDE = 27$ *units*,

$\therefore$ $ED + DC + EC = 27 \Rightarrow ED + 6 + 12 = 27 \Rightarrow ED = 27 - 18 = 9$ *units*

$\therefore$ $\frac{AB}{9} = \frac{8}{6} \Rightarrow AB = 12$ units and hence $AB = EC$.

Example 5.7 (Illustrating AA Similarity)

In the given Fig. 5.18,

if $\angle 1 \equiv \angle 3$ and $\angle 2 \equiv \angle 4$ then,

prove that $\Delta BIG \sim \Delta FAT$. Also find FA.

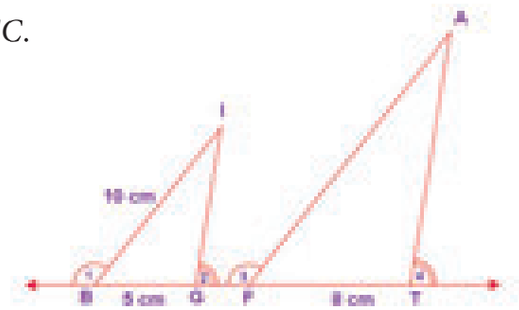


Fig. 5.18

Proof:

	Statements	Reasons
1	$\angle 1 \equiv \angle 3$	given
2	$\angle IBG \equiv \angle AFT$	supplements of congruent angles are congruent.
3	$\angle 2 \equiv \angle 4$	given
4	$\angle IGB \equiv \angle ATF$	supplement of congruent angles are congruent
5	$\Delta BIG \sim \Delta FAT$	by AA property (2, 4)

Also, their corresponding sides are proportional

$\Rightarrow \frac{BI}{FA} = \frac{BG}{FT} \Rightarrow \frac{10}{FA} = \frac{5}{8} \Rightarrow FA = \frac{10 \times 8}{5} = \frac{80}{5} = 16\,cm$

05_8th_Maths_Geometry_Chap 5.indd 163 2/12/2020 12:16:54 PM